AJINKYA AWARI

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Education

University College London (UCL), London, UK ***2026-2027***
MSc - Master of Science, Artificial Intelligence for Biomedicine and Healthcare

Savitribai Phule Pune University, Pune, IN | **CGPA: 8.25** ***2019-2023***
BE - Bachelor of Engineering, Computer Science
Leadership: Vice President - Technology Club

Skills

- **Languages :** Python, C, C++
- **ML / DL Frameworks :** PyTorch, TensorFlow, Keras, Scikit-learn
- **AI & Research Areas :** Graph Neural Networks, Transfer Learning, Reinforcement Learning, Neuro-symbolic AI, XAI, NLP / LLMs, Optimisation Theory
- **Data Science Tools and Technologies:** Hadoop, Apache Spark, SQL, BigQuery, Docker, Matplotlib
- **Web & Tools:** HTML, CSS, Bootstrap, Flask, Streamlit, Git, AWS, GCP

Experience

The Leadership 30 ***July 2024 – Present***
Senior Data Analyst

- Transforming Education in Maharashtra (SDG 4 - Quality Education): Led a data-driven initiative with the Maharashtra government on a three-month project to assess learning gaps in public schools. Developed targeted intervention strategies that boosted student engagement by 15% and improved literacy outcomes for 2,000+ students in rural districts.
- Advancing Clean Energy Access (SDG 7 - Affordable and Clean Energy): Designed a cost-benefit analysis model for renewable energy projects in rural Maharashtra, facilitating policy recommendations for solar micro grids. The initiative led to a 10% increase in clean energy adoption, benefiting 800+ households and reducing annual carbon emissions by 1,500 tons.
- Disaster Resilience & AI-Driven Risk Mitigation: Built an AI-powered flood forecasting system in collaboration with local agencies, enhancing emergency response times by 12% and safeguarding many individuals in 6 flood-prone districts across Maharashtra.

Byju's Think & Learn Pvt. Ltd, India
Cloud Engineer ***July 2023 – July 2024***

- Optimized cloud-based lead management automation by integrating LeadSquared, reducing manual effort by 4+ hours per week and boosting lead conversion rates by 10%, contributing to $300K+ in additional revenue. Utilized cloud monitoring and performance metrics to refine iterations of infrastructure and automation.
- Designed and optimized a scalable cloud infrastructure capable of handling 300,000+ concurrent users by improving backend architecture, database queries, and load balancing, reducing latency by 15% and ensuring 99.7% uptime during peak enrollments.
- Implemented a predictive analytics pipeline on cloud platforms using Python, enabling sales teams to prioritize high-potential leads, improving lead-to-customer conversion by 12% and driving an 8% increase in quarterly revenue.
- Automated cloud-based user onboarding workflows through Python-powered CRM integration, reducing onboarding time by 25%, minimizing errors by 30%, and enhancing user satisfaction by 15%, leading to lower churn rates.

Projects

Plant Disease Detection Using Machine Learning (Paper Published)

- Developed a plant disease detection system using CNN-based machine learning, published in IJARSCT Vol. 3, Apr 2023.
- Achieved 89% classification accuracy across multiple disease categories using image-based detection.
- Gained first experience in experimental design, model evaluation, and research documentation

ChestXplain: Explainable AI for Chest X-ray Classification

- Built a chest X-ray classification system using DenseNet121 with transfer learning on 20,000 NIH images across 14 pathologies, achieving mean AUC of 0.769
- Applied Grad-CAM analysis to examine how spatial feature preservation in dense blocks affects the quality of visual explanations
- Investigated the gap between statistical accuracy and clinical reliability, finding that architecture choices directly determine explanation trustworthiness.

GNN Agricultural Networks

- Compared GCN, GraphSAGE and GAT against an MLP baseline on node classification tasks across graphs ranging from 200 to 5,000 nodes
- GraphSAGE outperformed the graph-unaware MLP by 12.8 points in macro-F1, with the advantage peaking at 1,000 nodes
- Found that deeper GCN layers collapsed due to over-smoothing, and that residual connections only partially recovered performance, confirming the depth-aggregation tradeoff on sparse graphs